Dilemma Dynamics

Final evaluation: mathematical and engineering notes

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Abstract

The replacement final evaluation compares Direct and bounded Draft procedures in two hosted Nemotron models at temperatures 0 and 1. Its frozen selection rule chose 48 matched blocks per temperature: 7,104 episodes in total. The campaign is in progress. These notes specify the payoff references, estimands and evidence boundaries; they contain no scientific results. The maintained operational specification is `FINAL_EVALUATION.md` in the repository root.

1 What is being compared?

The two models use these exact IDs under the provider's `nvidia/` namespace:

- Nano: `NVIDIA-Nemotron-3-Nano-30B-A3B`
- Super: `nemotron-3-super-120b-a12b`

Direct makes one thinking-off action request, capped at 128 completion tokens. Draft makes a thinking-on request capped at 1,024 tokens, then a separate thinking-off action request capped at 128 tokens. The current draft conditions that action; earlier drafts are not carried forward. Each phase uses its episode's temperature and top-$p = 1$.

The treatment changes computation, request count, thinking controls and conditioning together. An observed difference would concern these bounded procedures, not an equal-cost isolation of reasoning. Generated drafts are not verified beliefs or faithful causal explanations of the selected action.

Players maximise expected own cumulative payoff. Decisions remain sealed until joint reveal; each player sees completed public history. Neutral X/Y labels are mapped to the following actions in both mappings and display seats:

Own action	Other cooperates	Other defects
Cooperate	$(3, 3)$	$(0, 5)$
Defect	$(5, 0)$	$(1, 1)$

Entries are (own payoff, other payoff). Cooperation is an outcome of interest, not the scoring objective. Against an always-cooperating opponent, defecting can improve payoff while reducing cooperation.

2 Committed opponents and exact references

Track A discloses the opponent's complete policy: AllC always cooperates; AllD always defects; Tit-for-Tat (TFT) cooperates first and then copies the subject's previous action; Grim cooperates until a subject defection, then defects forever. Each is tested at continuation probabilities $\delta \in \{.4, .6, .9\}$. One-round AllC/AllD and announced ten-round TFT/Grim give 16 conditions.

Let s encode the disclosed opponent's relevant memory, $r(s, a)$ be own payoff and $f(s, a)$ the next state after action a. In the geometric game,

$$Q^*(s, a) = r(s, a) + \delta V^*(f(s, a)),$$
$$V^*(s) = \max_{a \in \{\mathrm{C},\mathrm{D}\}} Q^*(s, a).$$

The implementation uses rational arithmetic and checks a candidate value function against every Bellman inequality. A certified best response to a committed policy is distinct from an equilibrium between two strategic players.

For a known finite horizon, use backward induction with $V_0^*(s) = 0$:

$$Q_h^*(s, a) = r(s, a) + V_{h-1}^*(f(s, a)), \qquad V_h^*(s) = \max_a Q_h^*(s, a).$$

Here h is rounds remaining, including the current round. In the announced ten-round TFT or Grim condition, nine cooperations followed by a final defection earn $9 \cdot 3 + 5 = 32$ own points and are optimal from the initial state. This conclusion uses the opponent's commitment, rather than a claim of rational cooperation between two unconstrained finite-game players.

Why these continuation probabilities?

For Grim's untriggered initial state, perpetual cooperation gives $3/(1-\delta)$; defecting immediately gives $5 + \delta/(1-\delta)$. Cooperation is optimal initially when $\delta \geq 1/2$.

For TFT, the profitable alternative is not always permanent defection. The cycle D,C against an initially cooperating TFT earns 5,0 repeatedly, with value $5/(1-\delta^2)$. Comparing this with $3/(1-\delta)$ gives the initial cooperation threshold $\delta \geq 2/3$; the engine verifies all alternatives and subsequent states. Equality permits multiple optimal actions. Thus .4, .6 and .9 span distinct incentive regimes.

The geometric stopping draw is hidden at decision time. Its expected length is $1/(1-\delta)$, so $\delta = .9$ means ten expected rounds, not an announced ten-round endpoint. Models cannot condition on the realised final round.

3 Loss and interaction outcomes

For each revealed committed-policy decision, define the nonnegative action gap

$$g_t = V^*(s_t) - Q^*(s_t, a_t),$$

using remaining-horizon values for a fixed game. Optimal continuation is allowed after the chosen action. The gap therefore measures the achievable expected own payoff given up at that decision, rather than comparing against a particular reference trajectory.

For a realised geometric episode of length T, or a fixed episode of length H, report

$$L_{\text{geometric}} = (1-\delta)\sum_{t=1}^{T} g_t, \qquad L_{\text{fixed}} = \frac{1}{H}\sum_{t=1}^{H} g_t.$$

No additional factor δ^{t-1} belongs in the realised geometric sum: termination already supplies survival weighting in expectation, and continuation is already included in Q^*. For example, C against AllC in a one-round game has gap $5-3=2$, while D has gap zero. Loss is in normalised payoff points, not percentage accuracy or cooperation rate.

Autonomous interaction is a different target

Track B tests all ten unordered pairings of the four model–procedure arms, including self-play, at $\delta = .9$. Neither agent receives an opponent policy. There is no identified exact best-response oracle for these adaptive opponents. The public analysis instead reports:

- first-round mutual cooperation and joint-action counts;
- mutual cooperation per observed round and per episode;
- normalised joint return, $(1-\delta)\sum_{t=1}^{T}(r_{1t}+r_{2t})$;
- breakdown and recovery counts with their opportunity denominators.

The joint return is not mean payoff per observed round. Long episodes contribute more to pooled round rates; rounds in the same dyad are dependent. A high cooperation rate does not by itself demonstrate equilibrium, robustness to exploitation, or superior own-payoff decisions.

Incomplete observations

Every scheduled episode has a status. Missing actions are not imputed as cooperation, defection or zero loss. Dynamic programming bounds feasible loss along the remaining scheduled length, conditional on any committed prefix. These finite-schedule bounds describe missingness exposure; they are not population confidence intervals. Complete-block estimates remain conditional on completion. An incomplete campaign cannot pass the release gate.

4 Replication, estimands and uncertainty

A fresh master seed fixes environment draws and randomised order. Each of 48 blocks per temperature contains 64 Track A episodes and ten Track B dyads. The resulting counts are 6,144 benchmark episodes and 960 autonomous dyads across both temperatures. Each of the four label/seat strata contains 12 blocks.

Corresponding arms and temperatures share the block's environment schedule; temperature counterparts are adjacent in randomised order. This pairing controls stopping and presentation differences and reduces time-window confounding. It does not match provider sampling noise or freeze backend weights.

Nested 24/32/48-block candidates were frozen before engineering. The largest candidate fitting measured token and latency forecasts was selected before scientific inference, using resource telemetry alone. The rule cannot read cooperation, correctness, loss or draft arguments. There is no outcome-based extension, favourable-seed search or superiority stopping rule.

Let $L_{m,p,\tau,b,c}$ be Track A episode loss for model m, procedure p, temperature τ, block b and condition c. Define

$$\bar{L}_{m,p,\tau,b} = \frac{1}{16} \sum_{c=1}^{16} L_{m,p,\tau,b,c}.$$

The four primary estimands are Draft minus Direct loss within each model and temperature, averaging blocks within strata and weighting the four strata equally:

$$\Delta_{m,\tau} = \frac{1}{4} \sum_{s=1}^{4} \frac{1}{|B_s|} \sum_{b \in B_s} (\bar{L}_{m,\mathrm{Draft},\tau,b} - \bar{L}_{m,\mathrm{Direct},\tau,b}).$$

Negative differences favour Draft. Conditions receive equal weight; long episodes do not dominate the benchmark merely by containing more rounds.

Use 50,000 deterministic PCG64 whole-block bootstrap draws, resampling within strata. Primary marginal intervals are 98.75%: a Bonferroni allocation of a 5% family across four contrasts, subject to approximate bootstrap coverage. At least five complete within-model blocks per stratum are required for an interval. Neither coverage nor power is guaranteed by this threshold.

Six secondary estimands compare temperature 0 minus 1 for each arm and the change in Draft-minus-Direct for each model. The same resampled block indices must be used jointly across temperatures. Secondary 95% intervals are nominal, not a multiplicity-controlled discovery family. Condition breakdowns, truncation, cost, latency and provider drift are exploratory; associations do not identify why drafting works or what a larger budget would cause.

5 Evidence, infrastructure and reproducibility

The existing local CPU coordinates remote Nebius Token Factory inference. There is no training, local model hosting, GPU allocation or hosted tracker. Inspect AI supplies native samples, model events and custom scored outcomes. An audited gateway owns HTTP transmission, reservations and recovery. Native joint-reveal events are linked to the canonical event chain by hashes.

MLflow Skinny records real local run identities, parameters, metrics and artifacts in SQLite. It is a reconstructable projection of the canonical ledger; it does not make model calls reproducible or replace raw evidence. NumPy provides the bootstrap, PyArrow the numeric Parquet exports and Matplotlib the primary figure. The lockfile and runtime manifest record dependency versions and installed state.

Each stage retains append-only request and reveal events, native Inspect logs, accounting, statuses and exports. The offline analysis produces episode histories, decision/request tables, condition means, contrasts and checksum manifests. Private responses and drafts stay outside the public package. Checksums detect changed bytes; they do not authenticate provider identity, billing or model weights.

Failure handling and limits

The additional campaign ceiling is $40: $2 engineering, $34 final inference, $2 other-charge reserve and $2 contingency. Retry exposure is included within stage allocations, not added above them. Reservations precede transmission; ambiguous requests retain their exposure. Durable valid responses are replayed, while invalid successful responses are not regenerated to obtain better outputs. The campaign deadline is 36 hours with a separate watchdog and shutdown margin.

Resume preserves seeds, schedule, prompts, claims, limits and original clock. An interrupted attempt is never quietly renamed into a new campaign. Earlier experiments motivated interleaving and bounded recovery, but are excluded from this replacement evaluation. Prior outcome inspection informed the design; this is not an outcome-naive first study.

Publication boundary

Only the completed verified replacement becomes the primary report and app evaluation. Both temperatures, native log coverage, resolved final reservations, MLflow export and reproducible derived artifacts must pass release checks before Results, replay and downloads switch together. Interim samples are not final claims. There is no established general model ranking, latent-belief validity, educational benefit or real-world negotiation performance.

Exact offline reconstruction from saved evidence is the reproducibility target. An independently repeated hosted request may differ even at temperature zero. The repository's `FINAL_EVALUATION.md` provides commands and concrete artifact names; `AGENTS.md` protects the running campaign's frozen inputs.

6 Separate app mathematics

The interactive app also exposes model forecasts and small competitive board games. These features are outside the scientific Direct/Draft treatment.

Forecasts and counterfactuals

When requested by the optional app protocol, forecasts are generated reports. Observed outcomes can be checked against those reports, but a reported probability is not direct access to a model's latent belief. Deterministic policy demos must not invent such telemetry. A one-step counterfactual holds the opponent's sealed current action fixed; it is not evidence that a different history would leave a strategic opponent unchanged. Verified game arithmetic and narrative interpretation should be displayed separately.

Tic-tac-toe

The small finite game admits exhaustive minimax. With utility expressed for the player to move, a terminal state's value is its game result, and nonterminal values satisfy the zero-sum recursion

$$V(s) = \max_{a \in A(s)} [-V(f(s, a))].$$

Draws have value zero. A legal losing action is distinct from a malformed move; search values, legality and model explanations should not be conflated.

Dots & Boxes

The state contains claimed edges, box ownership, current scores and the player to move. Claiming at least one box grants another move. Consequently a negamax recursion retains the sign when the same player moves again and reverses it when the turn passes. Exact search is practical only for sufficiently small remaining states; incomplete searches must be marked unknown.

Supported long-chain and loop references cover restricted endgames. A local box sacrifice can preserve control and improve final score, but a heuristic explanation is not an exact proof outside the reference's assumptions. These visible sequential games have competitive fixed-total scoring, so their tactical sacrifices do not establish the mutual-gain incentives of the repeated dilemma.

Implementation references

The Python package contains rational scoring and certification. The TypeScript game modules contain the interactive rules, search and analysis. Tests exercise known states and invariants without hosted inference. The original compatibility names `gamefield_research` and `gamefield-study` are retained to preserve execution and provenance.

Documentation for the underlying interfaces: Inspect evaluation logs; Inspect custom scorers; MLflow tracking.